

AHLAM S. ALQARNI

Applied Physics & Nanotechnology | Research & Development | Advanced Materials
Saudi Arabia | +966 593 511 262 | aalqarni2050@gmail.com

Objective

Applied Physics and Nanotechnology researcher specializing in magnetic materials, thin films, and advanced functional nanomaterials with peer-reviewed international publications. Strong laboratory expertise in magnetic and transport measurements and structural characterization. Research supports thermal management technologies, magnetic devices, and advanced materials systems relevant to high-performance industrial and defense applications.

Core Technical Expertise

Magnetic materials and ferromagnetic thin films; Magnetocaloric effect (MCE) materials; Functional alloys and nanostructured materials; Magnetic and transport measurements; X-ray diffraction (XRD) analysis; Experimental materials evaluation and scientific data analysis.

Research Highlight

Discovered for the first time a new Magnetocaloric Effect behavior in engineered magnetic thin films, contributing to advancements in thermal switching and next-generation thermal management technologies for compact and efficient cooling applications.

Education

Ph.D. Applied Physics & Nanotechnology – The Catholic University of America, USA (2025)
M.Sc. Applied Physics & Nanotechnology – The Catholic University of America, USA (2021)
M.Sc. Physics – Morgan State University, USA (2018)
B.Sc. Physics – King Khalid University, Saudi Arabia

Research Experience

Materials Physics Researcher – The Catholic University of America, USA: Magnetic and transport characterization of advanced thin films; study of intermetallic and ferromagnetic materials.
Laboratory Research Assistant – Morgan State University, USA: Thin film preparation and structural and magnetic measurements.

Publications

A. Lisfi, S. Pokharel, **Ahlam S. Alqarni**, et al. "Static magnetism and thermal switching in randomly oriented L10 FePt thin films." AIP Advances, 8, 056438 (2018).

Subhadeep Datta, **Ahlam S. Alqarni**, et al. "Large intrinsic anomalous Hall effect in near-gapless half-metallic ferromagnet Fe₂Si thin film." Journal of Alloys and Compounds, 1055, 186446 (2026).

Additional research papers are currently under peer review and will be published soon.

Technical Tools

XRD | Magnetic measurements | Transport measurements | Thin film analysis | Scientific data processing

Languages

Arabic (Native) | English (Professional)